Faculty of Science and Arts
Department of Chemistry
Biochemistry Lab (CHEM 266)

Experiment 3

SEPARATION OF AMINO ACIDS BY PAPER CHROMATOGRAPHY

Pre-Lab Questions:

1. In Figure 1b, number the spotted amino acids and arrange them according to their polarity, descending.
2. What does it means if you see a big spot after you develop your chromatogram?
3. Why should we immerse the solvent at a level lower than the pencil line that we draw during doing the paper chromatography experiment?
4. Why should we use a pencil to draw the starting line? Why we can't use a ballpoint pen?
5. What would happen if you forget to cover the container that contains your chromatogram and the solvent?
6. Does R_f has a unit?

Precautions:

1. You are going to use a hot plate to heat your chromatogram in order to develop a colored spot, be carful not to touch it!
2. Don't touch the middle of paper directly with your fingers; fingerprints will affect the results of your experiment.

Objectives:

1. Define chromatography.
2. Explore the concept of paper chromatography
3. Be able to calculate the retention factor R_f for different amino acids and figure out the unknown amino acids from the calculated values

Introduction:

Chromatography (Greek: *Chromos* meaning color, and *graphein* meaning to write) is a technique that is discovered by a Russian-Italian botanist Mikhail Tsweet. Mikhail Tsweet was the first scientist who separated the plant pigments using solid adsorbents.

Chromatography is a term often employed to refer to a group of analytical and preparative techniques, it is used to separate, purify, and identify a mixture of chemical compounds or biomolecules. There are two phases in this technique, the stationary phase and the mobile phase. The separation is mainly dependent on the deferential adsorption of the various components of the mixture to the stationary phase. Chromatography is used to separate, purify, and identify substances.

Paper chromatography is one of the most common types of chromatography. In paper chromatography the stationary phase is a cellulose or filter paper over which an organic solvents passes (mobile phase).

In paper chromatography, the sample mixture is spotted onto a base line marked by pencil only at the cellulose or the filter paper; next, the edge of the paper is immersed in a solvent, which is usually a mixture of water and an organic liquid. The solvent is then carried up by the capillary action holding the components of the mixture. In general, the paper which is composed from the cellulose (a highly polar polymers), adsorbs the more polar molecules compared to less-polar solvent. Therefore, we expect that the polar molecules to adsorb strongly on the filter paper thus move a shorter distance compared to non-polar molecules in a mixture of chemical compounds (figure 1).

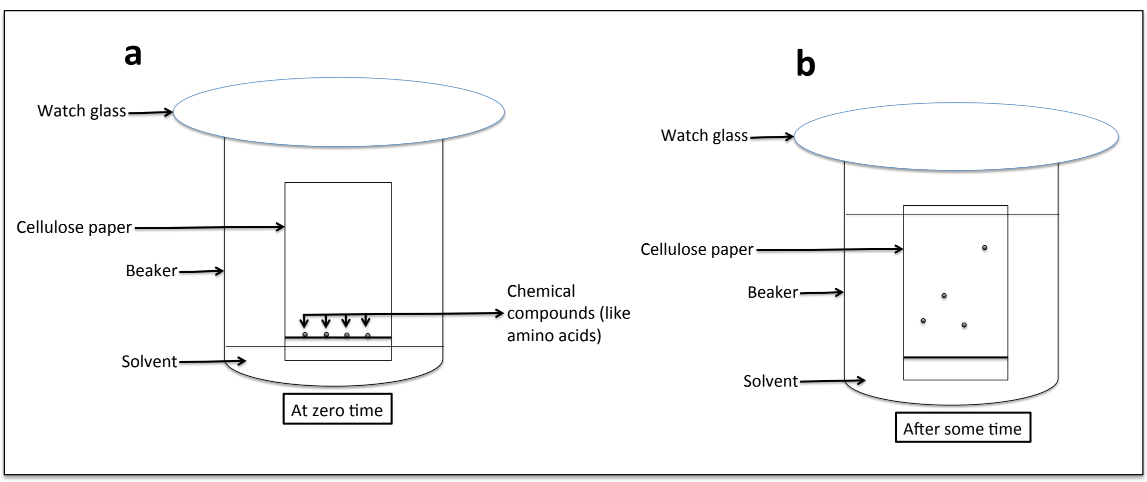

Figure 1. The general set up of paper chromatography experiment. **a)** Different amino acids are spotted ~2 cm away from the bottom, then the paper is immersed in a solvent, which will travel up by the capillary action. **b)** After some time, the solvent traveled up through the filter paper, move up the applied amino acids at variable distance because they have different polarities, therefore they adsorb differently on the filter paper.

In order to distinguish a specific chemical in any paper chromatography experiment, R_f (Retention Factor or Ratio to front) is calculated. R_f is defined as the distance traveled by the chemical to the distance traveled by the solvent (figure 2).

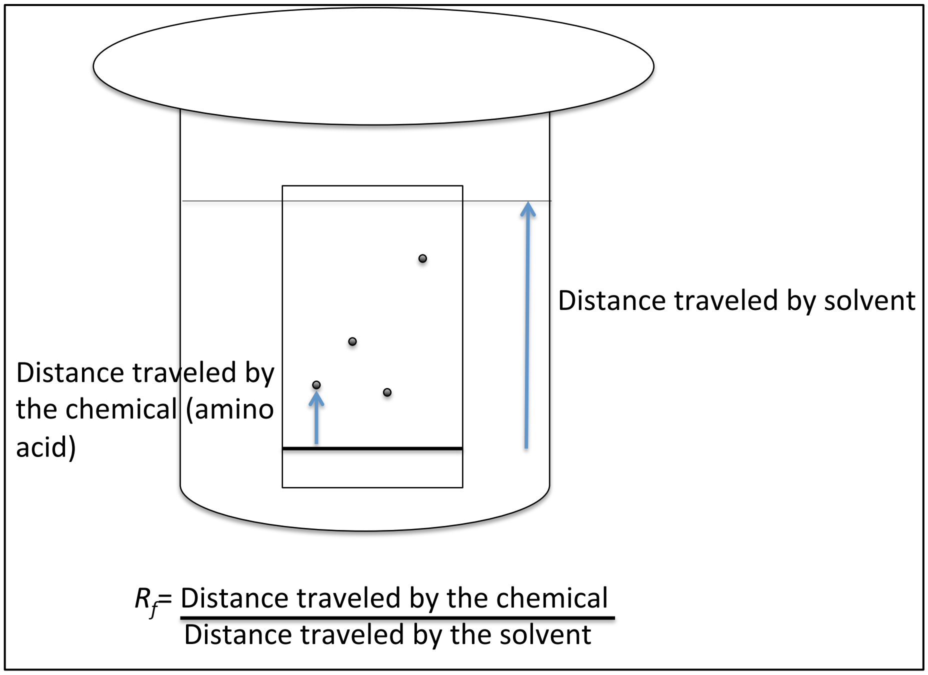

Figure 2. Retention Factor, Ratio to front, or R_f, formula.

In order to be able to calculate the R_f, of the spots (in our experiment, the amino acids spots) on the chromatogram are colored by performing the Ninhydrin test, where purple or yellow color is developed. Please back to the qualitative analysis of amino acid experiment.

Experimental procedure:

<u>Materials needed:</u>

1. 0.5M amino acid solutions
2. Developing solvents:
   a) 20 µl ammonia in 10 ml distilled water, shake and take out 5 ml of this solution in 100 ml distilled water and 400 ml methanol.
   b) A mixture of butanol: acetic acid: distilled water in the ratio of 12: 3: 5 respectively.
3. Ninhydrin solution that is freshly prepared; 50 mg ninhydrin in 25 ml acetone.
4. Filter paper sheets.

<u>Procedure:</u>

1. Place a reasonable amount of developing solvent in chromatographic jar and cover about 10 minutes, why?

2. On reasonable rectangular filter paper, draw a line about 2 cm from its edge, you have to use pencil line.
3. From the left side of the filter paper, mark 6 pencil dots on the line.
4. Spot the amino acid samples by dipping a fine capillary tube in the acid solution and just touch each dot by a particular acid or acid mixture.
5. Allow the spots to dry, stand the spotted paper in the covered chromatographic jar in a way that the base line is above the level of the developing solvent.
6. When the developing solvent reaches the front line or after ~40 minutes, take out the chromatogram.
   Note: If the developing solvent doesn't reach the front line within the 40 minutes and you stop the process of chromatograph, you should have to draw a new front line at the top of developing solvent before praying it with ninhydrin.
7. Dry your chromatogram then spray it with ninhydrin.
8. Heat on a hot plate or in an oven at about 110 ^{0}C until the colored spots appears clearly.
9. Calculate the R_f and identify the unknown.

Data analysis:

1. Write the report of this experiment in your awn words.
2. Upon starting the chromatography experiment, be aware not to touch the paper directly with your fingers, please wear your gloves. Apply small amount of the amino acid on each spot that you made by the pencil and on your notebook write the name of each amino acid to remember its position.
3. When you calculate the R_f for each amino acid, please indicate clearly the formula and the unit of each parameter in the formula, DON'T write just anumbers!
4. Identify the unknowns and explain, in your own words how did you figure that out.

References:

1. Introductory organic and biochemistry laboratory manual, Kansas State University, revised 2009.

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Separation of Amino Acids by paper chromatography

Report Author:

Lab Partner:

Date Due:

Date Turned In:

Unknown #: **Unknown Identification:**

Purpose:

Data and calculations:

Please attach your chromatogram

Amino Acid	Retardation Factor (R_f)
Alanine	
Aspartic acid	
Arginine	
Proline	
Glutamic acid	
Glycine	
Unknown #, Name:	

Conclusion:

Errors and problems you encountered and their possible remedies: